Further results on the brief *Physical vector spaces for quantum mechanics*

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Abstract

This note collects results that go beyond the technical brief *Physical vector spaces for quantum mechanics* [1] and partial answers to its open problems. They are not part of the brief. Each section is self-contained, and each names the Lean declarations that check it; the Lean files are in the brief's directory, under `QuantumVectorSpaces/FurtherResults/`.

1 The Hilbert brief's change of frame is not a map of Schwartz space

The brief proposes quantum vector spaces (QSTO:V) as a replacement for the Hilbert-space formulation QSTO:H, and argues in the proof of its Theorem 5 that QSTO:V satisfies [B]WDQ. The argument assumes that "the defining observables of A must be observables in B and vice-versa" for two frames A and B of the same system. This premise holds exactly when the two frames give the same topology on V (checked with the brief, `QuantumVectorSpaces.topology_eq_iff_continuous`). It is not automatic. This section shows that it fails for the brief's own Example 3, $V = \mathcal{S}(\mathbb{R})$, with the change of frame that the companion brief *Unphysicality of Hilbert spaces for quantum mechanics* [2] uses against QSTO:H.

That change of frame is $y = \tan(\frac{\pi}{2}\operatorname{erf} x)$. Its unitary U on $L^2(\mathbb{R})$ sends the Gaussian packet

$$\psi(x) = \left(e^{-x^2}/\sqrt{\pi}\right)^{1/2} = \pi^{-1/4}e^{-x^2/2} \quad \text{to} \quad \phi(y) = \left(\pi(1+y^2)\right)^{-1/2}.$$

Theorem 1.1. *1. ψ is a Schwartz function, and $U\psi = \phi$ is not. So U does not map $\mathcal{S}(\mathbb{R})$ into itself.*

2. The position Y of the second frame, carried back to the first as $U^{-1}YU$, is not defined at the Schwartz function ψ.

Proof. Every Schwartz function f lies in the domain of the position operator, since xf is again a Schwartz function and so in L^2. But $\int y^2|\phi(y)|^2\,dy = \int y^2/(\pi(1+y^2))\,dy = \infty$, so ϕ is not in that domain (the brief *Unphysicality of Hilbert spaces*, Proposition 3). This gives (1). For (2), $\psi \in D(U^{-1}YU)$ iff $U\psi \in D(Y)$, which fails. □

So the two observers do not describe the same V with two families of defining observables, as the premise requires: the second observer cannot express the first observer's ψ in $\mathcal{S}(\mathbb{R})$. QSTO:V gives [B]WDQ only for changes of frame whose unitaries map V onto itself. The restriction is stated in print by Carcassi, Calderón and Aidala: on the Schwartz space "the only unitary transformations we are interested in are those that map Schwartz functions to Schwartz functions", and the change

of coordinate "is automatically ruled out" [3, Appendix, Sec. VIII]. The brief does not repeat it. Theorem 1.1 is the concrete instance, checked in Lean.

Which changes of variable do map $\mathcal{S}(\mathbb{R})$ into itself is known: Galbis and Jordá characterize them [4, Thm. 2.3]. That the restriction cannot be lifted by choosing a different space is shown in the companion note on the Hilbert brief [9, Sec. 2]: no nonzero state of $L^2(\mathbb{R})$ has finite position variance in every smooth frame both now and after a free evolution. A quantum vector space has to come with its group of changes of frame (Sec. 3).

1.1 Lean

Module:
`QuantumVectorSpaces.FurtherResults.SchwartzFrame`,
namespace `QuantumVectorSpaces.FurtherResults`. The unitary U (`frameUnitary`), the packets and the position operator are those of the Hilbert brief's formalization.

- `toLp_gaussSchwartz`: ψ is $\pi^{-1/4}$ times TauCeti's Gaussian Schwartz map.
- `toLp_mem_positionDomain`: every Schwartz function lies in the domain of the position operator on $L^2(\mathbb{R})$.
- `frameUnitary_gaussState_not_mem_range`, `not_forall_frameUnitary_mem_range`: Theorem 1.1(1).
- `toLp_gaussSchwartz_not_mem_domain`: Theorem 1.1(2).

The statement is made in $L^2(\mathbb{R})$, where U lives; it does not need U to be smooth.

2 The completeness problem of Definition 1

The brief leaves one problem with its Definition 1 open (Sec. 5, "Completeness of the Lie algebra"). The definition admits two cases that the brief calls pathological: "a Hilbert space with the defining observables of a qubit", which has "too many vectors", and "a Schwartz space with only position in the defining observable", which has "too few observables". This section shows that the second case is not a quantum vector space at all, and that the first one is excluded by an irreducibility condition which Examples 3 and 4 of the brief satisfy.

Throughout, V carries an inner product and defining observables X_i, and $\mathcal{A}$ is the unital algebra they generate. The topology of the brief's Corollary 2 is given by the seminorms $p_A(v) = \|Av\|$, $A \in \mathcal{A}$; Definition 1 requires V to be complete and second countable in it.

2.1 Position alone

Theorem 2.1. *Let $V = \mathcal{S}(\mathbb{R})$ with the L^2 inner product, and let position X be the only defining observable. Then V is not complete in the topology of Corollary 2. So it is not a quantum vector space, whatever $\hbar$ and structure constants are chosen.*

Proof. Let $\psi_0(x) = \pi^{-1/4} e^{-x^2/2}$ and $f_m = u_m \psi_0$ with $u_m(x) = \sqrt{(m+1)^{-2} + x^2}$, a Schwartz function since u_m is smooth with polynomially bounded derivatives. As $u_m(x) = |(m+1)^{-1} + ix|$, the triangle inequality gives $|u_m - u_k| \le |(m+1)^{-1} - (k+1)^{-1}|$ and $|u_m(x) - |x|| \le (m+1)^{-1}$. Every $A \in \mathcal{A}$ commutes with multiplication by u_m, so

$$p_A(f_m - f_k) = \|(u_m - u_k)\, A\psi_0\| \le \left|(m+1)^{-1} - (k+1)^{-1}\right| \|A\psi_0\|,$$

and (f_m) is a Cauchy sequence. If it converged to some $h \in \mathcal{S}(\mathbb{R})$, then $\|f_m - h\| \to 0$ (take $A = 1$), while $\|f_m - |x|\psi_0\| \leq \|\psi_0\|/(m+1)$. So $h = |x|\psi_0$ almost everywhere, and everywhere since both are continuous. But $|x|\psi_0$ is not differentiable at 0. □

So the brief's second example is excluded by Definition 1 already, through completeness. The pathology it was meant to show does not go away, though. The completion of V is the space Φ_X of all $f \in L^2(\mathbb{R})$ with $x^k f \in L^2(\mathbb{R})$ for every k, and Φ_X with position alone is a quantum vector space: it is a separable Fréchet space, the projective limit of the weighted spaces $L^2((1+x^2)^k\,dx)$. Neither statement is formalized. It still has "too few observables": multiplication by e^{ix} commutes with position, on Φ_X as on $\mathcal{S}(\mathbb{R})$.

2.2 Irreducibility

Definition 2.2. V with the defining observables X_i is *irreducible* if every linear map $T\colon V \to V$ that is continuous in the topology of Corollary 2 and commutes with every X_i is a multiple of the identity.

This is a form of the condition Roberts imposes on labelled Hilbert spaces: "We simply suppose that $\mathcal{O}' = \mathcal{O}'_l$, where $\mathcal{O}'$ denotes the set of all bounded operators commuting with $\mathcal{O}$" [5, Sec. 2, p. 1099], where $\mathcal{O}$ is the set of all observables and $\mathcal{O}_l$ that of the labelled ones. Without superselection rules $\mathcal{O}' = \mathbb{C}1$, and the condition says that the labelled observables have only the scalars in their commutant. Definition 2.2 states this on V itself, with continuous instead of bounded operators, and so needs no Hilbert completion. For unbounded operators "commutant" can be made precise in several inequivalent ways; the statements below are for this one. A sufficient condition for continuity is easy to check: if T commutes with every X_i, it commutes with every $A \in \mathcal{A}$, and $\|Tv\| \leq C\|v\|$ for all v gives $p_A(Tv) = \|TAv\| \leq C\,p_A(v)$.

Theorem 2.3. *Let $\hbar \neq 0$.*

1. *Example 4 of the brief, $\mathbb{C}^2$ with the spin observables $S_i = \frac{\hbar}{2}\sigma_i$, is irreducible.*
2. *$\mathbb{C}^2 \otimes \mathbb{C}^2$ with the defining observables $S_i \otimes 1$ is a quantum vector space, with the same commutation relations, but it is not irreducible: $1 \otimes \sigma_3$ commutes with every $S_i \otimes 1$.*
3. *$\mathcal{S}(\mathbb{R})$ with position alone is not irreducible: multiplication by e^{ix} commutes with position.*
4. *Example 3 of the brief, $\mathcal{S}(\mathbb{R})$ with X, $P = -i\hbar\,d/dx$ and I, is irreducible.*

Proof. (1) A 2×2 matrix that commutes with σ_3 is diagonal, and a diagonal one that commutes with σ_1 is scalar. (2) The commutation relations and symmetry pass from S_i to $S_i \otimes 1$, and the remaining conditions hold in every finite dimension; finite dimension also makes $1 \otimes \sigma_3$ continuous. (3) Multiplication by e^{ix} is an isometry for the L^2 norm, hence continuous by the remark above, and it is not scalar.

(4) The ladder operators $a_\mp = X \pm d/dx$ lie in $\mathcal{A}$, since $d/dx = (i/\hbar)P$. On the Hermite functions ψ_n they act by $a_-\psi_n = \sqrt{2n}\,\psi_{n-1}$ and $a_+\psi_n = \sqrt{2(n+1)}\,\psi_{n+1}$. Write $c_n(f) = \langle \psi_n, f\rangle$ and $S_m f = \sum_{n<m} c_n(f)\psi_n$. The partial sums converge to f in $\mathcal{S}(\mathbb{R})$, because the coefficients of a Schwartz function decrease rapidly [8]. Since X is symmetric and d/dx antisymmetric, $c_n(a_- g) = \langle a_+\psi_n, g\rangle = \sqrt{2(n+1)}\,c_{n+1}(g)$; so $a_- g = 0$ forces $c_n(g) = 0$ for $n \geq 1$, and $g = c_0(g)\psi_0$. Now let T be continuous and commute with X and P. It commutes with $a_\mp$, so $a_- T\psi_0 = T a_-\psi_0 = 0$ and $T\psi_0 = z\psi_0$ for some z. Applying a_+ gives $T\psi_n = z\psi_n$ for every n by induction, hence $TS_m f = zS_m f$, and in the limit $Tf = zf$. □

Parts (1) and (2) also bear on the brief's claim, in the proof of its Theorem 5, that "two systems are physically equivalent if and only if their Lie algebras of observables are equivalent". The two spaces have isomorphic Lie algebras of defining observables and the same structure constants, but they are not isomorphic, as their dimensions differ. Irreducibility alone does not restore the claim: spin $\frac{1}{2}$ and spin 1 are both irreducible, with the same commutation relations. The system is determined by the representation, not by the Lie algebra; for the Heisenberg algebra the Stone–von Neumann theorem makes the irreducible representation unique once $\hbar$ is fixed.

2.3 The group form

A stronger version of the same repair is to make a group part of the data. Let G be a Lie group, π an irreducible unitary representation on a Hilbert space H, and let V be the space H^∞ of smooth vectors, with the seminorms $\|d\pi(u)v\|$ for u in the enveloping algebra. These are the seminorms of Corollary 2 for the defining observables $i\,d\pi(Y_k)$, with (Y_k) a basis of the Lie algebra. Antoine builds his space from such a representation and requires it to be invariant under the symmetry group [6]; he considers H^∞ and sets it aside because it need not be nuclear. For the Schrödinger representation of the Heisenberg group, H^∞ is $\mathcal{S}(\mathbb{R})$ with its usual topology [7, Secs. 1 and 6]. In this form every defining observable is essentially self-adjoint on V, as Antoine recalls from Segal and from Nelson and Stinespring, and it generates a unitary group that maps V onto itself. Every $\pi(g)$ maps V onto itself, which is the condition on changes of frame that Sec. 1 found missing from the brief's argument for [B]WDQ. The cost is that the definition presupposes the Hilbert space and the group. The Lean files contain only the Lie-algebra form above; Theorem 2.3(4) is its instance for the Heisenberg group.

2.4 Lean

Module:
`QuantumVectorSpaces.FurtherResults.Irreducibility`,
namespace `QuantumVectorSpaces.FurtherResults`.

- `not_completeSpace_posOnly`, `not_isQuantumVectorSpace_posOnly`: Theorem 2.1.
- `IsIrreducible`: Definition 2.2; `continuous_of_commute`: the sufficient condition for continuity.
- `isIrreducible_qubit`, `isQuantumVectorSpace_pair`, `not_isIrreducible_pair`, `not_isIrreducible_posOnly`, `isIrreducible_schwartz`: Theorem 2.3(1)–(4).
- The convergence of Hermite expansions in $\mathcal{S}(\mathbb{R})$ is `SchwartzMap.tendsto_hermiteSum`, in `ForMathlib.Analysis.Distribution.SchwartzSpace.HermitePartialSum`.

Not formalized: that Φ_X is a quantum vector space, the remark on Theorem 5, and the group form.

3 [B]WDQ relative to a group of frames

Section 1 and the companion note [9, Sec. 2] show that [B]WDQ cannot hold for "all frames" as the brief words it: some changes of frame do not map V into itself, and no nonzero state serves every smooth frame at once. This section states [B]WDQ with the frames as part of the data and shows that the brief's argument then goes through, with a condition that can be checked.

Definition 3.1. A *frame group* for V is a group F with a homomorphism ρ from F to the linear bijections of V such that every $\rho(g)$ preserves the inner product. The frame g has the defining observables $X_i^g = \rho(g)^{-1} X_i \rho(g)$: the observer in frame g describes the state v by $\rho(g)v$ and measures X_i on it. The frame group satisfies *[B]WDQ* if for all $g, h \in F$ the defining observables of g are observables of h, that is, symmetric operators continuous in the topology that Corollary 2 gives to the X_i^h.

The last condition is the premise of the brief's proof of Theorem 5, for every pair of frames in F. The frame 1 is the original one.

Theorem 3.2. *Let the X_i be symmetric and let (F, ρ) be a frame group.*

1. *(F, ρ) satisfies [B]WDQ if and only if every frame gives the same topology on V, and if and only if every $\rho(g)$ is continuous, that is, F acts by automorphisms of the topological vector space V.*

2. *Then all frames have the same observables, and every $\rho(g)$ is a unitary equivalence of V with itself. So it carries the pure states, with their topology, the Born probabilities and the observables of one frame to those of another.*

3. *A sufficient condition is that every X_i^g lies in the algebra $\mathcal{A}$ generated by the X_i.*

4. *The translations $f \mapsto f(\cdot - a)$, $a \in \mathbb{R}$, are a frame group of Example 3 that satisfies [B]WDQ, for every $\hbar$.*

5. *The change of frame $y = \tan(\frac{\pi}{2} \operatorname{erf} x)$ belongs to no frame group of Example 3: no linear map of $\mathcal{S}(\mathbb{R})$ agrees with its unitary U.*

Proof. (1) Since $\rho(g)$ preserves the inner product, the topology of the frame g is the topology of the frame 1 pulled back along $\rho(g)$; this is the transport result already used for isomorphisms (checked with the brief, `QuantumVectorSpaces.induced_topology_eq_pullback`). Conjugation by $\rho(g)$ also preserves symmetry. By the brief's frame criterion (`QuantumVectorSpaces.topology_eq_iff_continuous`), [B]WDQ for the pairs $(g, 1)$ and $(1, g)$ says that the topology of the frame g equals that of the frame 1, and conversely equal topologies give [B]WDQ for every pair. A topology t equals its pullback along $\rho(g)$ for every g exactly when every $\rho(g)$ is continuous for t: continuity of $\rho(g)$ gives one inclusion, and continuity of $\rho(g^{-1}) = \rho(g)^{-1}$ gives the other. (2) follows from (1). (3) If every X_i^g lies in $\mathcal{A}$, then $X_i = \rho(g)^{-1} X_i^{g^{-1}} \rho(g)$ lies in the algebra of the frame g, and two frames whose defining observables lie in each other's algebras give the same topology. (4) Under translation by a, position becomes $X + a$ and momentum stays P, both in $\mathcal{A}$; the inner product is preserved because Lebesgue measure is translation invariant. (5) is Theorem 1.1(1). □

So the brief's argument for [B]WDQ is correct once "all frames" is read as "all frames of a given frame group that acts by automorphisms of V". Part (1) says that this restriction is also necessary. Antoine requires the same invariance of his space under the symmetry group [6, Sec. II], and his Proposition 4, on observables that transform by a finite-dimensional representation, is related to part (3). The negative half is not in his paper: part (5), and the companion note's result that no space works for all smooth frames. Which frame group a quantum vector space should carry is a choice made with the space. For $\mathcal{S}(\mathbb{R})$ the changes of variable that preserve it are those of Galbis and Jordá [4, Thm. 2.3], with the half-density factor added for unitarity; in the group form of Sec. 2, F is the group G itself.

3.1 Lean

Module:
`QuantumVectorSpaces.FurtherResults.FrameGroups`,
namespace `QuantumVectorSpaces.FurtherResults`.

- `FrameGroup`, `frameObs`, `IsFrameCompatible`: Definition 3.1.
- `isFrameCompatible_iff_topology_eq`, `isFrameCompatible_iff_continuous`: Theorem 3.2(1).
- `IsFrameCompatible.isObservable_iff`, `IsFrameCompatible.unitaryEquiv`: Theorem 3.2(2); the pure states and observables are carried by the brief's `UnitaryEquiv.pureStates` and `UnitaryEquiv.isObservable_conj_iff`.
- `isFrameCompatible_of_mem_algebra`: Theorem 3.2(3).
- `translate`, `frameObs_translate`, `isFrameCompatible_translate`: Theorem 3.2(4).
- `not_exists_frameUnitary`: Theorem 3.2(5).

References

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[3] G. Carcassi, F. Calderón, C. A. Aidala, The unphysicality of Hilbert spaces, Quantum Stud.: Math. Found. **12**, 13 (2025), arXiv:2308.06669v3.

[4] A. Galbis, E. Jordá, Composition operators on the Schwartz space, Rev. Mat. Iberoam. **34** (2018) 397–412.

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